PREPARED FOR NRG | EXECUTIVE WORKING DISCUSSION

PROTECTING TIME TO POWER

A working model for integrated electrical delivery across NRG's generation buildout.

PRESERVE UPSTREAM CERTAINTY THROUGH THE ELECTRICAL DELIVERY CHAIN.

Stephan Hardt | Director, Power Generation Solutions | Southwire

WHY NOW

NRG's growth agenda creates three distinct electrical-delivery environments.

1.5 GW

TEF PORTFOLIO

One project operating; two targeting mid-2028.

5.4 GW

NEWBUILD DEVELOPMENT

Turbine and EPC access supports a runway through 2032.

25.8 GW

OPERATING FLEET

Approximately 1-2 GW of upgrades under evaluation.

DIFFERENT ASSET STATES. ONE REQUIREMENT.

Preserve schedule certainty through energization.

Specifications, capacity, releases, reels, package interfaces and field readiness still have to converge on COD.

Source: NRG 2Q26, Aug. 4, 2026

THREE RISKS

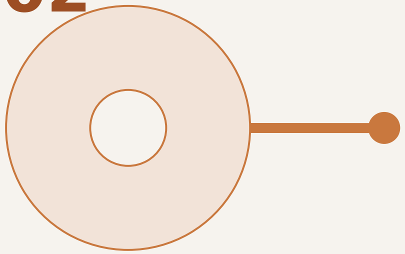
Three execution risks can erode certainty between design and energization.

01

LABOR PRESSURE

Electrical labor demand can peak when schedule-recovery and commissioning flexibility are lowest.

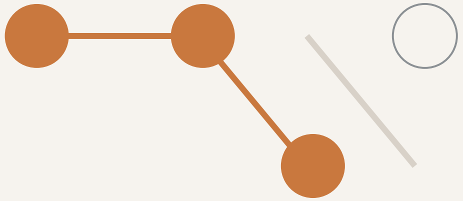
02



MATERIAL COMPLEXITY

Specifications, lengths, reels, accessories and releases must match the construction sequence.

03



FRAGMENTED COORDINATION

Owner, EPC, OEM, supplier and contractor decisions can converge too late.

LATE HANDOFFS BECOME SCHEDULE AND COMMISSIONING EXPOSURE.

Labor availability | material readiness | decision timing

TIME TO POWER IS AN ELECTRICAL EXECUTION PROBLEM BEFORE IT IS A CABLE-BUYING PROBLEM.

- Interfaces cross packages, zones, contractors and turnover paths.
- Late material decisions return as handling, rework and commissioning exposure.
- Managing the workstream as a system can protect schedule better than isolated purchase-order optimization.

DESIGN — MATERIAL — WORKFACE — ENERGIZATION

ONE COORDINATED FLOW

A coordinated electrical delivery system aligns decisions from planning through turnover.

Southwire connects application support, material planning, sequenced logistics and field readiness—without changing project accountability.

1 **PLAN + FORECAST**
Scope and capacity visibility

2 **APPLICATION ALIGNMENT**
Codes, duty and alternates

3 **MATERIAL + REEL STRATEGY**
Routes, cuts and reel plan

4 **RELEASE + LOGISTICS**
Sequence by zone and milestone

5 **INSTALLATION READINESS**
Labels, kits and pull planning

6 **FIELD SUPPORT + TURNOVER**
Feedback, testing and records

SOUTHWIRE COORDINATION RAIL Shared data | governed handoffs | field feedback

NRG governance, EPC/EOR/OEM design authority and contractor means and methods remain intact.

FIELD-ENGINEER WHAT IS UNIQUE. PREFABRICATE WHAT REPEATS.

THE OUTCOME

The value to NRG is fewer execution surprises—not more supplier services.

Translate each delivery lever into a direct execution outcome.

ALIGN APPLICATIONS + INTERFACES EARLY	→	Fewer late changes and handoff failures
GATE CAPACITY + RELEASES	→	More predictable material availability and sequencing
PLAN REELS + REPEATABLE SETS TO THE WORKFACE	→	Less handling and avoidable rework
CARRY FIELD RECORDS THROUGH TURNOVER	→	Cleaner commissioning and lifecycle continuity

OUTCOME HYPOTHESES TO VALIDATE ON ONE LIVE NRG SCOPE.

ILLUSTRATIVE ADJACENT-MARKET BENCHMARK | NOT AN NRG RESULT

Adjacent experience indicates where to test—not what NRG will save.

Field hours per electrical spine in one repetitive modular-generation scope

4,400

OBSERVED FIELD HOURS
SITE-BUILT BASELINE

→

~880

MODELED FIELD HOURS
WITH FACTORY-BUILT ASSEMBLY

WHAT IT SUGGESTS

Stable, repeatable interfaces may be candidates for offsite work.

WHAT IT DOES NOT PROVE

Net NRG savings, schedule reduction or critical-path removal.

Rebuild the case with NRG scope, labor rates, schedule logic and acceptance criteria.

ONE LIVE SCOPE

One live scope can prove—or disprove—the case before NRG expands it.

Start with one active newbuild package, TEF project or acquired-fleet conversion/uprate screen.

SUGGESTED FIRST SCOPE: **CEDAR BAYOU 5 — ACTIVE TEF UNIT, MID-2028 COD** · OUR SUGGESTION, YOUR CALL · DETAIL IN APPENDIX A9

01 — **SELECT SCOPE**
One active electrical package

02 — **WALK THE WORK**
People closest to execution

03 — **MAP THE RISK**
Handoffs, releases and repetition

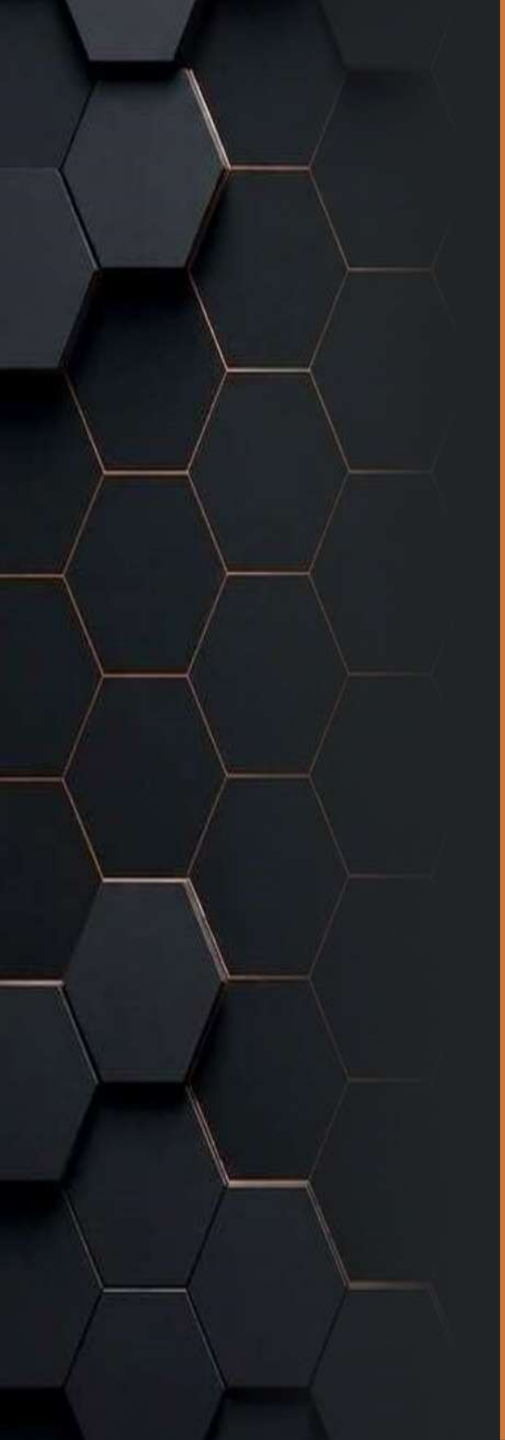
04 — **DECIDE**
Stop, standardize or pilot

WRITTEN READ-BACK

- Milestone exposure
- Release + reel opportunities
- Repeatable-interface candidates
- Role gaps + recommendation

STOP
STANDARDIZE
PILOT

No presumption that the answer is prefab—or that Southwire belongs in the solution.



THE ASK

ONE SITE WALK.
ONE HOUR.
NO SALES PRESENTATION.

Choose one live electrical scope and bring the people closest to execution.

INPUT	**One live scope**
OUTPUT	**Written execution-risk read-back**
DECISION	**Stop, standardize or define a bounded pilot**

NO PRICING EXERCISE. NO BROAD COMMERCIAL COMMITMENT.

Stephan Hardt | Director, Power Generation Solutions | Southwire

stephan.hardt@southwire.com | +1 470-439-8488

SOURCE DISCIPLINE

NRG context is current, attributed and non-additive.

1.5 GW TEF PORTFOLIO	One operating project and two projects targeting mid-2028. Do not describe the entire portfolio as under construction.
5.4 GW NEWBUILD DEVELOPMENT	Turbine and EPC access supports a runway through 2032. Do not call all 5.4 GW committed or under construction.
25.8 GW OPERATING FLEET	NRG's reported operating-generation fleet after the LS Power acquisition.
~1-2 GW UPGRADE POTENTIAL	CT-to-CCGT conversions and traditional uprates are under evaluation—not a committed pipeline.

DO NOT SUM THESE FIGURES: THEY REPRESENT DIFFERENT ASSET STATES AND OVERLAP CONCEPTUALLY.

DECISION RIGHTS

Clear decision rights preserve existing project accountability.

NRG	Owner priorities, portfolio standards, risk tolerances, acceptance criteria and investment decisions.
EPC / EOR	Design authority, calculations, specifications, routing basis and approved technical design.
OEM	Equipment-package interfaces, internal wiring, approved terminations and warranty boundaries.
ELECTRICAL CONTRACTOR	Installation means and methods, pull planning, workface execution, testing support and field feedback.
SOUTHWIRE	Application support, material and capacity visibility, engineered reel strategy, sequenced delivery and selected field support.

Southwire connects selected decisions; it does not replace owner, EPC, OEM or contractor accountability.

PLANT-ZONE COVERAGE

Six system clusters capture plant-wide electrical complexity.

1, 16

GRID + SITE BACKBONE

Switchyard, GSU/grid tie and MV corridors
HV/EHV · MV · protection/control · fiber · grounding

2, 3, 10

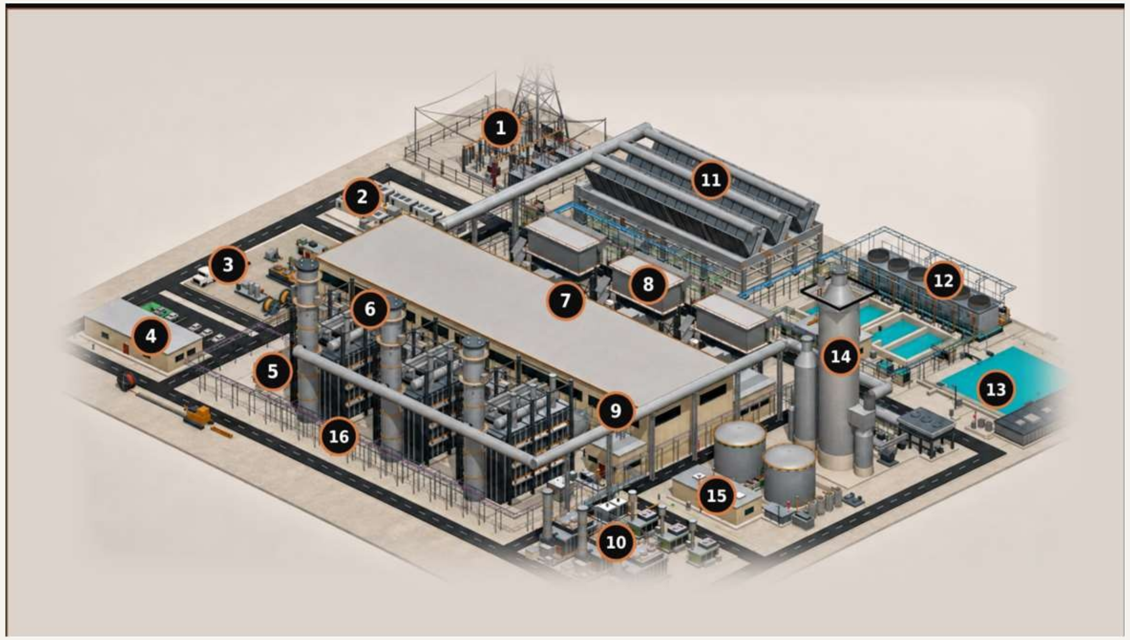
MODULAR + LOGISTICS

BESS, reel staging, prefab/skids and modular power
DC/MV/LV · controls · fiber · engineered sets

4, 5, 9

BUILDINGS + DISTRIBUTION

Control building, e-house and MCC/VFD/UPS
MV/LV/VFD · essential AC/DC · life safety · BAS/data



6, 7, 8

GENERATION ISLAND

HRSG, turbine hall and GT inlet
MV/LV/VFD · I&C · F&G/ESD · CEMS · heat trace

11, 12, 13

HEAT REJECTION + WATER

ACC, cooling and water/wastewater
Repeated fan/pump power · VFD · controls · wet runs

14, 15

OPTIONAL + AUXILIARY

Carbon capture and gas metering
Classified power/control · instrumentation · F&G · fiber

CABLE CLASSES HV/EHV | MV 5-35 kV | LV 600 V | VFD | essential AC/DC | I&C | fiber/network | life safety | grounding

Boundary: generator-output bus, OEM-internal wiring and contractor means and methods remain excluded.

DEFINED SUPPORT

Southwire can support selected delivery stages without taking design authority.

ALIGN

Application and BOM review
Alternative-construction evaluation
Multidisciplinary cable support

COORDINATE

Project cable management
Inventory and release coordination
Reel, cut and sequence planning

PREPARE + SUPPORT

Striping, marking and labeling
Kitting and bundling
Selected field and lifecycle support

PROCUREMENT + SUSTAINABILITY SUPPORT

Product-level EPDs and responsible-production evidence can support procurement where applicable to the selected product and manufacturing site.

Capability does not equal outcome. Availability, geography, contractual scope and technical fit must be confirmed for each project.

METHODOLOGY BEFORE CLAIMS

The adjacent benchmark is useful only when its boundaries are visible.

OBSERVED BASIS

- Define the site-built scope and the unit called one spine.
- Confirm time records, sample size, project conditions and exclusions.
- Document what remained in field labor, supervision and commissioning.

MODELED BASIS

- Reconcile scope equivalence between observed and modeled work.
- Include engineering, factory work, freight, FAT and field installation.
- Test whether any activity actually leaves the critical path.

DO NOT CLAIM SAVINGS UNTIL THE GROSS-TO-NET CASE IS REBUILT WITH NRG INPUTS.

Release gate: methodology, scope equivalence and external-use permission approved.

THE OWNER'S EQUATION

Cable is a small line item with large coefficients.

LCOE and IRR decide every project. Five variables actually move — none of them is the purchase price.

LCOE — COST PER MWh

$$\text{LCOE} = \frac{\text{CAPEX + O\&M + Fuel (lifetime, discounted)}}{\text{Lifetime MWh delivered (discounted)}}$$

Lower the top. Grow the bottom. De-risk the rate that discounts both.

IRR — RETURN ON THE PROJECT

$$NPV = \Sigma\, CF_t \div (1 + IRR)^t = 0$$

Most sensitive to COD date, capex certainty and availability — revenue that starts late is IRR you never get back.

01 SCHEDULE → COD

Cable sits on the energization critical path. Supply certainty and sequencing protect the revenue start date — the largest single IRR lever.

02 RISK → DISCOUNT RATE

Evidence-based scope lowers contingency and financing risk: an estimate within 4% on footage was 26% off on copper — right in total, wrong where the money is.

03 CAPEX — THE NUMERATOR

Cable is under 2% of plant capex — purchase price barely moves the top line. Installed cost does: routing, reels, pulls, rework.

04 MWh — THE DENOMINATOR

Availability and asset life grow the bottom of the equation: cable health, rejuvenation and life extension keep MWh flowing.

05 COMMODITY VOLATILITY

Copper and aluminum exposure managed through supply strategy and hedging — capex certainty that lenders can underwrite.

CABLE BARELY MOVES THE NUMERATOR — IT MOVES COD, RISK AND AVAILABILITY, WHERE LCOE AND IRR ARE DECIDED.

ROUTED 3×1 NGCC MODEL

The site model behind A3 is quantified, not conceptual.

Every figure below comes from the routed Rev 89 geometry — built before any RFQ existed.

96 SCHEDULE RECORDS	NGCC master cable schedule Rev M21 — geometry-governed quantities, not factored estimates.
1,784 CIRCUITS ROUTED	2,478 physical runs routed through the Rev 89 site geometry, across every plant zone in A3.
1,136,255 PROCUREMENT FT	Approximately 1.14M ft of cable derived from routed lengths plus engineering allowances.
563,091 LB COPPER	About 0.50 lb of copper per foot on average — commodity exposure quantified before procurement.

Published Southwire record: 15M+ ft of HV underground cable with zero in-service failures.

THE MODEL BEHIND APPENDIX A3 — CONCEPT, NOT FOR CONSTRUCTION; STOCK NUMBERS REQUIRE CONDUCTOR SIZING.

Southwire
POWER GENERATION SOLUTIONS

PLANT ZONES & KEY CABLE APPLICATIONS

WHAT EACH ZONE POWERS — AND WHY THE CABLE CHOICE MATTERS.

APPENDIX A

SK-120

3x1 NGCC · REV 89 SITE MODEL · CONCEPT, NOT FOR CONSTRUCTION

1 SWITCHYARD, GSU BANK & GRID TIE

Overhead gen-tie conductors or HV/EHV underground cable, plus protection/control, fiber and grounding. POI boundaries, accessories and testing can gate energization.

ACSR/ACSS • HV/EHV XLPE • TC-ER control • fiber • bare copper

2 BESS YARD

Flexible BESS DC cable, MV collection, LV auxiliaries, controls and fiber. Repeated containers and PCS skids support a standardized DC/LV/control/fiber spine; MV collection follows the plant one-line.

RenewaFLEX 2 kV • DLO • MV-105 • XHHW-2/TC-ER • fiber

3 REEL STAGING, LAYDOWN & PREFABS / SKIDS

Zone-tagged reels, engineered cut lengths, spine sections and prefab / skid cable sets span all cable classes. Installation-sequenced staging reduces splices, scrap, handling and congestion.

MV-105 • XHHW-2 • TC-ER • VFD cable • MC-HL

4 ADMIN BUILDING & CONTROL ROOM

Low-voltage fire alarm, access control, HVAC/BAS and data join building power here. Separation protects life-safety and controls.

THHN/THWN-2 • Genesis fire alarm • Genesis access control • Genesis HVAC/BAS • fiber/data

5 E-HOUSE — MODULAR ELECTRICAL BUILDING

MV feeders, LV/VFD power, controls, fiber, station DC and panel wire. Defined E-house interfaces make it the spine head-end, with pre-engineered cable sets reducing field terminations.

MV-105 • XHHW-2/TC-ER • shielded VFD • PLTC/ITC • SIS

6 HRSG TRAINS & STACKS

Motor/valve power, lighting/small power, heat trace, thermocouple/RTD, CEMS and F&G/ESD cable serve HRSG skids. Pre-engineered harnesses simplify interfaces; heat and vertical runs still govern supports.

XHHW-2/TC-ER • MC-HL • PLTC/ITC F&G/ESD • thermocouple/RTD • heat-trace

7 TURBINE HALL — GAS & STEAM TURBINES

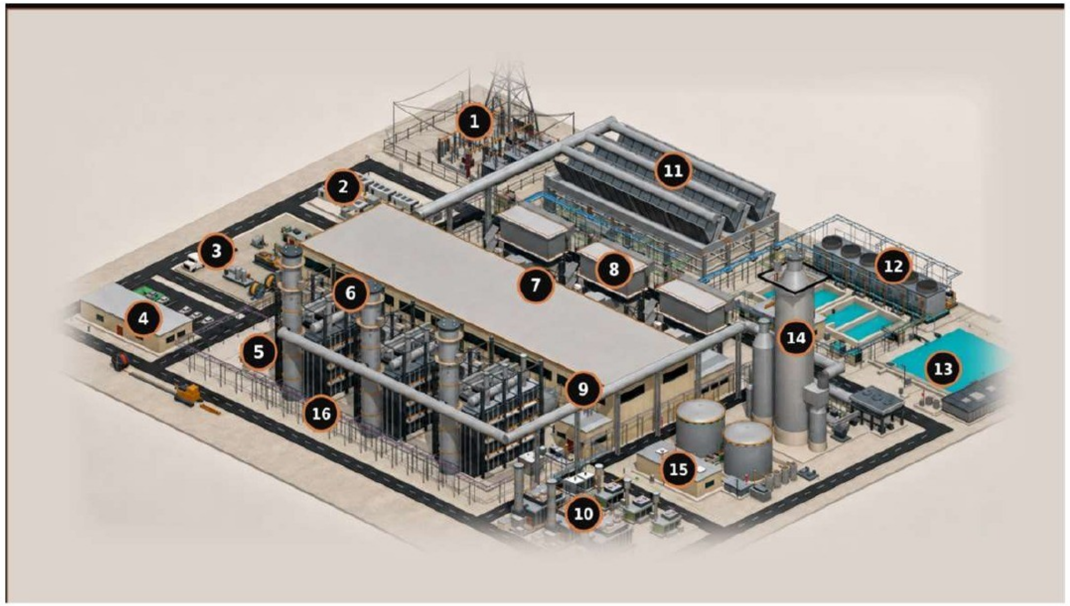
MV/LV/VFD power, generator/turbine controls, F&G/ESD, lighting/small power and festoon cable serve turbine-hall skids. A skid spine simplifies auxiliaries; generator output remains isolated-phase bus.

MV-105 • ARMOR-X MV VFD • XHHW-2/TC-ER • PLTC/ITC F&G/ESD • festoon

8 GT AIR INLET SECTION

Fan and heater power, controls, plus shielded pressure and temperature instrumentation. Repeated inlet houses support standardized drop assemblies tied to a common control interface.

TC-ER • shielded VFD • PLTC/ITC • thermocouple/RTD • fiber



CABLE CLASSES CARRIED BY THE SCHEDULE

Bus duct and the overhead getaway are boundary only — excluded from all totals

- MV 5–35 kV
- LV 600 V
- I&C / instrument
- DC 125 V
- Essential AC
- Fibre / network
- Grounding, bare

HV UNDERGROUND 69–500 kV • MV EPR 133% 5 / 15 / 25 / 35 kV • DUCT BANK, DIRECT BURIED AND TRAY

Published Southwire record: 15M+ ft HV underground, zero in-service failures

96	1,784	2,478	1,136,255	563,091	~0.50
SCHEDULE RECORDS	CIRCUITS ROUTED	PHYSICAL RUNS	PROCUREMENT FT	LB COPPER	LB CU / FT AVG

SCHEDULE BASIS

QUANTITIES FROM THE NGCC MASTER CABLE SCHEDULE REV M21, GEOMETRY-GOVERNED. SPECIFICATION NUMBERS ARE SOUTHWIRE FAMILY REFERENCES — A STOCK NUMBER REQUIRES CONDUCTOR SIZING AND IS DELIBERATELY NOT SHOWN. ZONE NUMBERS ARE KEYED TO THIS VIEW.

9 MCC, MOTOR VFD & UPS ROOM

LV/MV motor feeders, VFD cable, essential AC/DC and fire-alarm/control CI wiring originate here where required. Outgoing sets simplify pulls; shielding and grounding still manage PWM stress and EMI.

MV-105 VFD • ARMOR-X VFD • Machine Flex VFD • DLO 2 kV • Circuit Defender

10 MODULAR POWER YARD — GENSETS, SIMPLE CYCLE, BTM, BLACK START, FUEL CELLS

MV/LV, control, fiber and grounding can be coordinated into prefabricated electrical spines for modular packages and skids. Consolidated terminations reduce field pulls and speed installation.

MV-105 • XHHW-2/TC-ER • DLO/RHH/RHW-2 • PLTC/ITC • fiber

11 ACC — AIR-COOLED CONDENSER

VFD power, controls, vibration/temperature instrumentation and fiber serve repeated fan cells. One trunk-and-drop electrical spine can repeat bay by bay across the ACC.

shielded VFD • XHHW-2/TC-ER • PLTC/ITC • vibration/RTD • fiber

12 CHILLERS & COOLING TOWERS

Chiller/pump/tower feeders, MV/LV VFD cable, HVAC/BAS controls and submersible cable serve packaged skids and cells. Modular spines speed installation; wet runs remain field-engineered.

MV-105 • ARMOR-X MV VFD • XHHW-2/TC-ER • PLTC/ITC HVAC/BAS • submersible pump

13 WATER TREATMENT, PROCESS TANKS & WASTEWATER POND

Pump/VFD power, TC-ER cable, MC-HL where classified, shielded instrumentation and submersible cable. Dosing and pump skids can use prewired power/control/instrument whips; wet-pit and submersible runs stay field-engineered.

XHHW-2/TC-ER • shielded VFD • MC-HL • PLTC/ITC • submersible

14 CCS — ABSORBER / REGENERATOR

LV/MV feeders, VFD cable, MC-HL, instrumentation, F&G/ESD, heat trace and fiber serve CCS packages. If OEM modules and skid interfaces are frozen, they can form a repeatable CCS spine; reserve corridor and tray capacity early.

MV-105 • ARMOR-X MV VFD • MC-HL • PLTC/ITC F&G/ESD • heat-trace

15 GAS METERING / METER HOUSE

One umbilical combines metering-skid power, F&G/ESD and instrumentation. Heat trace and sealing protect accuracy.

MC-HL power/control • ARMOR-X instrumentation • PLTC/ITC F&G/ESD • heat-trace • fiber

16 SITE CABLE CORRIDOR & MV DISTRIBUTION

MV backbone, LV auxiliaries, lighting/small power, controls, fiber and grounding cross site corridors. Route geometry, thermal design and pull limits shape cable and reel strategy.

MV-105 • XHHW-2 lighting/small power • TC-ER control • single-mode fiber • bare copper

FASTER TIME TO POWER. LOWER EXECUTION RISK.
ENGINEERED FOR LONG-TERM RELIABILITY.

stephan.hardt@southwire.com | +1 470-439-8488
Stephan Hardt - Power Generation Solutions

SK-120 · 3×1 NGCC · Rev 89 site model · 16 zones with per-zone cable applications · concept, not for construction · print at 34" for site-walk use.

CEDAR BAYOU UNIT 5 · BAYTOWN, TEXAS (CHAMBERS COUNTY)

Cedar Bayou 5 is the natural first live scope.

A 721 MW combined-cycle brownfield expansion at NRG's existing Cedar Bayou station — the anchor TEF unit in the NRG / GE Vernova / Kiewit framework, in active construction toward a mid-2028 COD.

PROJECT FACTS — VALIDATED AUG 2026

721 MW CCGT · BROWNFIELD

GE Vernova H-class train at the existing station; ERCOT Houston Load Zone; existing 345 kV switchyard, cooling water and site infrastructure.

$562M TEF LOAN · 3% · 20-YEAR

Signed September 2025 — 60% of the $936M project cost; part of $1.15B TEF financing plus $66M tax abatements across the three-unit program.

COD TARGET MID-2028

First TEF unit to advance construction; T.H. Wharton (415 MW) completes mid-2026, Greens Bayou 6 follows in 2028.

EPC: TIC / KIEWIT · OEM: GE VERNOVA

Delivered under the 5.4 GW development framework — 1.2 GW in service by 2029, 1.2 GW by 2030, 3 GW more through 2032.

POTENTIAL SCOPE SELECTIONS

- HV generator leads and brownfield 345 kV switchyard tie-in for the H-class train
- MV plant auxiliary distribution, 5–35 kV, across the new unit and interfaces
- LV 600 V power and control — turbine hall, HRSG and balance of plant
- I&C, DCS and fiber — integration with the existing station and ERCOT SCADA
- Grounding-grid extension into the heritage plant grid
- Sequenced reels, cut lengths and kits against 2026–2028 procurement windows
- Post-COD continuity — records, spares and MRO carried into the operating fleet

Validated against public sources, Aug 2026: PUCT / Texas Energy Fund announcements, Utility Dive, NRG investor materials. Capacity reported 689–721 MW across sources; 721 MW per the PUCT loan announcement.

IN ACTIVE PROCUREMENT NOW — THE OBVIOUS CANDIDATE FOR THE ONE-LIVE-SCOPE VALIDATION PATH ON SLIDE 8.